

Submit the following five exercises. You are expected to provide full solutions in a linear and coherent fashion. You are encouraged to work in groups, but your work must be your own and must be self-contained.

Exercise 1.

Find all real numbers x satisfying $\ln(-x + 1) + \ln(6) = e$.

Exercise 2.

Consider the function $f(x) = e^{x \ln(x+2)}$.

1. Find the domain of $f(x)$.
2. Find all real numbers x satisfying $f(x) = 1$.

Exercise 3.

Find all real numbers x satisfying $e^{2x} + e^x - 6 = 0$.

Exercise 4.

Consider the function $f(x) = 6(7)^x$. Find c and k satisfying $f(x) = ce^{kt}$.

Exercise 5.

The population of a city is (roughly) modeled by the equation $P(t) = P_0(1 + r)^t$, where $P(t)$ is the population at time t , P_0 is the population at time $t = 0$, and $0 \leq r \leq 1$ is the annual growth rate. In 2020, the population of Metro Vancouver in 2020 was 3 million people and its rate of growth was 4.3%.

1. Find c and k such that $P(t) = ce^{kt}$.
2. Calculate the time it takes for the population to double.